Responsibility-Carrying State: Auditable Sufficiency, Reopen Contracts, and Safe State Reuse

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Abstract

A compact state is sufficient only relative to a downstream responsibility. We formalize this conditional notion and a certificate interface whose declared support controls reuse and reopening. An earlier benchmark reported zero unsafe reuse, but hostile audit showed that its harm endpoint had zero reachable opportunities; that empirical safety claim is therefore withheld rather than rescored. A prospectively frozen 30-case panel instead grades decisions against certificate-independent gold. In each of four corruption worlds, all 30 mutation opportunities are live: authenticated certificates reject every mutation, produce no unsafe reuse, and cost 0.6111 times always-raw recovery on the valid-certificate panel. A composed finite world then evaluates 12,288 episodes with 2,457 scheduled corruptions. The authenticated policy rejects all scheduled corruptions, makes no unsafe reuse, attains verified correctness 0.97933, and costs 0.539 times always raw overall; the unverified policy makes 330 unsafe reuses and 123 adversary-induced unnecessary reopens. The historical self-scored zero, two unsafe comparator arms, and a failed predecessor gate are all retained. These controlled finite-world results support responsibility-indexed reuse and authentication as a load-bearing mechanism; they do not establish external or population-level safety.

1 Introduction

Compression, abstraction and state reuse are normally evaluated against a named objective. Problems appear when compact state is reused after the downstream responsibility changes. A summary sufficient for a factual question may omit provenance needed to authorize a scientific claim. A proof-state abstraction useful for next-tactic prediction may omit dependencies needed to diagnose or repair a failed proof. A control-state abstraction adequate for one policy may collapse distinctions needed for intervention or counterfactual analysis.

This is not merely an uncertainty problem. A system can be highly confident while state omits a variable that becomes decisive under a new responsibility. Nor is it merely provenance: knowing exactly where state came from does not establish what that state can safely support.

ORION-23 asks:

> **Can a compact state carry a machine-checkable responsibility contract that says what it supports, what it omits, what changes invalidate reuse, and whether richer state can be reopened—without forcing every decision to reload raw evidence?**

The paper contributes responsibility-relative sufficiency; a responsibility-carrying state interface; permanent negative-result analysis; and an outcome-contingency adjudication that separates the exact supported core from descriptive P13A measurements whose empirical safety–cost authority is withheld.

2 Donor boundary and responsibility-relative sufficiency

Statistical sufficiency, state abstraction, bisimulation, predictive-state representations and causal abstractions already establish that different tasks can require different retained distinctions. Selective prediction and uncertainty gating use confidence to abstain. Provenance/evidence tracing binds artifacts to origin. Proof-carrying systems attach verifiable certificates. Memory-staleness work asks when stored information is no longer valid. ORION-23 claims none of these primitives.

The residual is **responsibility-scoped certified reuse with explicit reopen semantics and exact responsibility-support conditions**. Empirical safety–cost superiority remains unestablished because P13A's published harm endpoint was self-scored and had no reachable opportunities.

Let raw world `X` induce correct output `g_rho(X)` for responsibility `rho`, and let compact representation be `Z=T(X)`.

2.1 Exact responsibility sufficiency

`Z` is sufficient for responsibility `rho` over world set `Omega` if there exists `h_rho` such that `g_rho(x)=h_rho(T(x))` for every `x` in `Omega`. Equivalently, every equivalence class induced by `T` must be homogeneous under `g_rho`.

2.2 Responsibility-shift witness

A pair `(x,x')` witnesses insufficiency after `rho_L->rho_H` when `T(x)=T(x')`, the lower-responsibility outputs agree, and the higher-responsibility outputs differ. No learner is needed to establish such a witness; it is a property of the abstraction and responsibility.

For empirical systems, operational sufficiency debt is the verified higher-responsibility performance gap between richer and compact state, conditional on a prospectively frozen lower-responsibility equivalence/noninferiority requirement. It is a benchmark quantity, not a universal information measure.

3 Permanent negative history

The historical experiment constructed independent binary latent variables `(x,m,r)` and responsibilities `PREDICT=x`, `DECIDE=x`, `INTERVENE=x*m`, `VERIFY=x*m`, and `REPAIR=r`. Representations were `Z1=(x)`, `Z2=(x,m)` and `Z3=(x,m,r)`.

Exact enumeration produced the intended responsibility ladder: all representations were perfect for prediction/decision; `Z1` was exactly 0.5 on intervene/verify while `Z2/Z3` were 1.0; `Z2` was 0.5 on repair while `Z3` was 1.0. Exact upward debts were therefore +0.50.

However, the frozen protocol also required the maximum deviation among 100 finite-sample sanity replicates of `n=1024` to be ≤ 0.05. The observed maximum was `0.0556640625` at replicate 92. The combined terminal remains permanently `P14_CONTROLLED_SUFFICIENCY_DEBT_GATE_NOT_MET`.

3.1 Root cause

The harness grouped a finite sample by compact state, selected the majority target value from that same sample and credited that majority on the same observations, then maximized deviation over 100 replicates. The statistic combines within-sample majority optimism with an extreme-value operation. The correct response is not to change `0.05` after observing `0.0556640625`. The old terminal stays negative. The successor changes the estimand: exact support is evaluated by exhaustive equivalence classes, while efficacy is evaluated on a fresh safety–cost benchmark.

4 ResponsibilityCarryingState contract

An RCS contains compact state plus a fail-closed contract binding raw/source evidence identity; compiler/transform identity and version; exact supported responsibility set; independent witness/certificate identity; intentionally omitted coordinates/information classes; required-same context coordinates; reopen-on-change coordinates; raw recovery/reconstruction availability and freshness; recovery/reopen cost; resource envelope; evaluator identity; authority owner; and an explicit declaration that the object grants no scientific/novelty self-authority.

At reuse time the contract returns:

- `USE_COMPILED` when requested responsibility is supported and all bound conditions hold;
- `REOPEN_REQUIRED` when support does not hold but richer state can be recovered;
- `CANNOT_CHECK` when support is absent and recovery/verification cannot establish a safe route.

`CANNOT_CHECK` is a correct scientific disposition, not an error to hide.

5 Independent protected benchmark

The successor retains the interpretable world but evaluates support exactly: `Z1` supports {`PREDICT,DECIDE`}; `Z2` supports {`PREDICT,DECIDE,INTERVENE,VERIFY`}; and `Z3` supports all five responsibilities. The exact equivalence-class matrix is checked directly and matches the registered support sets.

Protected seed: `2026082113`. Families: **24**. Episodes per family: **512**. Total protected episodes: **12,288**.

Family probabilities for hidden variables `m` and `r` vary independently over `[0.65,0.95]`. Compact state is `Z1` or `Z2` with equal probability; requested responsibility is uniform over five tasks. Raw recovery is independently available with probability `0.95`. When compact state lacks a needed coordinate, an unqualified decoder uses the family MAP value, deliberately creating cases that can be high-confidence yet structurally unsupported.

5.1 Baselines

1. `UNQUALIFIED`: always reuse compact state.
2. `CONFIDENCE_ONLY`: reuse if estimated task accuracy is at least `0.80`; otherwise reopen when possible.
3. `PROVENANCE_ONLY`: valid lineage is present, so provenance alone permits reuse.
4. `RCS`: reuse only when exact responsibility support holds; otherwise reopen if raw is recoverable, else `CANNOT_CHECK`.
5. `ALWAYS_RAW`: reopen raw state whenever possible.

Fixed resource units are `REUSE=1`, `REOPEN=6`, `CANNOT_CHECK=0.5`.

Baselines 1 and 3 are the same policy on this corpus, and the list is four distinct policies rather than five. Every episode supplies valid lineage by construction, so the provenance check never refuses and `PROVENANCE_ONLY` reduces to "always reuse" — which is `UNQUALIFIED`. Their measured rates are identical on every metric reported: unsafe reuse `0.3961588541666667`, verified correctness

0.9248046875, mean cost 1.0. This follows from the construction above rather than from anything that happened during the run, and the results report the two together for that reason. It has one consequence worth stating plainly: this benchmark does not test whether a provenance check helps, because the check is never exercised. Establishing that would need episodes with absent or broken lineage, which this corpus does not contain.

6 Results

The historical runner printed its preregistered safety-cost terminal. The later outcome-entailment adjudication withholds that empirical claim authority. The current bounded authority comes from P13B's authenticated-certificate controlled finite-world result. Historical protected terminal: `P13A_RCS_SAFETY_COST_SUPERIORITY_SUPPORTED`. Current authority: `P13A_EMPIRICAL_SAFETY_COST_AUTHORITY_WITHHELD` under `P13A_OUTCOME_ENTAILMENT_ADJUDICATION_V1.json`.

arm	unsafe reuse	verified correctness	unnecessary reopen	mean cost
RCS	**0.0000**	**0.9807**	**0.0000**	**2.8747**
confidence only	0.2156	0.9657	lower than RCS	1.8582
provenance only	0.3962	0.9248	0	1.0000
unqualified compact	0.3962	0.9248	0	1.0000
always raw	0.0000	0.9513	0.5744	5.7319

RCS emits `CANNOT_CHECK` for all **237** certificate-declared unsupported/nonrecoverable cases and no other protected case. These action, rate, cost and replay numbers are descriptive historical facts. They do not establish elimination of unsafe reuse because the harm endpoint is self-scored.

6.1 Outcome-contingency adjudication

On the exact 3,840-point audit space, a registered certificate corruption moves the RCS action on 2,304 points but moves the published unsafe-reuse endpoint on zero. The endpoint has zero opportunities and returns `CANNOT_CHECK`. A control that grades the same reuse decisions against independently defined gold support has 1,536 live opportunities and passes, establishing that the missing denominator is specific to P13A's scorer rather than inevitable.

6.2 Why provenance fails

Every compact state has valid lineage. `PROVENANCE_ONLY` and `UNQUALIFIED` are identical policies and identical result rows here, so path or label multiplicity cannot create two independent comparisons.

6.3 Why always raw is not the answer

Always reopening has the reported historical P13A cost/action profile. P13A cannot establish the safety coordinate because its scorer was not independently gradable.

6.4 Certificate-independent finite panel

The panel defines gold support from task requirements and state variables without reading the certificate. The complete panel contains six state forms and five tasks. Omitted, overbroad, forged and stale-epoch worlds each have 30 live mutation opportunities before scoring. Authenticated RCS rejects every mutated certificate and makes zero gold-scored unsafe reuses in every world;

unverified RCS has 0, 5, 5 and 16 unsafe reuses in the omitted, overbroad, forged and stale worlds. On valid certificates, authenticated RCS is exactly correct and costs 0.6111 times always raw. Two fresh subprocess payloads are byte-identical, supporting the controlled finite-world claim.

6.5 Composed safety and efficacy world

The composed evaluation contains 12,288 episodes and 2,457 scheduled certificate corruptions. Authenticated reuse rejects every scheduled corruption and makes zero unsafe reuses. Its verified correctness is 0.97933, compared with 0.95247 for always-raw recovery and 0.98063 for reuse without certificate authentication. Its overall cost is 0.539 times always raw and its parent-form subpanel cost is 0.498 times always raw.

The similar correctness of the unverified arm does not make authentication decorative. Without authentication, the same world contains 330 unsafe reuses (66 forged, 87 overbroad and 177 stale) and 123 adversary-induced unnecessary reopens under omitted support. The contrast is finite and specification-bound; it is not a population estimate or external validation.

7 Certificate transport, related work and reproducibility

Responsibility support is conditional on evidence identity, transform version, required context, witness identity and resource envelope. A semantic change can require preserve, reopen, revoke or an explicit indeterminate disposition. The RCS object does not certify its own scientific authority. An evaluator can establish operational support; that role grants no novelty, publication or deployment authority.

Two donor boundaries are explicit. Truth-maintenance and provenance systems own the broad ideas of dependency-aware retraction and lineage. State-abstraction and predictive-state work own the broad idea that a representation may be sufficient for a task family. The present residual is narrower: sufficiency is indexed by a named responsibility, encoded in an authenticated certificate, and coupled to a material-reopen rule when that responsibility or its witness changes. Recovery cost is only the controlled study's pricing layer; the endpoint is the non-compensatory unsafe-reuse contract.

Recent proof-carrying action work attaches model-agnostic certificates to agent actions and runtime governance. Commit-time authorization, semantic transactions and stateful authorization-to-effect protocols bind authority, recovery and durable effects at later lifecycle boundaries. Provenance traces evidence and execution, while memory-staleness systems detect that stored state is no longer valid. These donors make it insufficient to claim simply that "state should carry a certificate." The discriminator here is the **responsibility key plus reopen semantics**: a state may be current, well-provenanced and high-confidence yet insufficient for another downstream responsibility.

The support matrix is deterministic and exact. The historical negative is reported exactly as frozen and is not reanalysed into a positive result. The certificate-independent study is a complete registered 30-case finite panel, not a population sample, and reports exact counts rather than a post-hoc inferential interval. Each corruption world must have a nonzero opportunity denominator before zero violations can be interpreted. A real-system extension should define task/domain as the unit of generalization and use paired matched comparisons with family/domain-block uncertainty.

8 Limitations and conclusion

Responsibilities are discrete and known in the controlled world; real systems may have ambiguous or compositional responsibilities. Exact sufficiency is strong; approximate support needs frozen tolerances and calibrated witnesses. The contract is only as reliable as external witness and recovery metadata. Resource costs are controlled units, not measured real latency/tokens/IO. High-risk domains may rationally choose always-raw. The paper has not yet demonstrated verifier-backed Lean repair/diagnosis or blinded scientific-workflow responsibility shifts, and certificate transport/revocation under real semantic version changes remains an external validation gate.

Sufficiency is a contract over future responsibility, not an intrinsic property of compact representation. The study preserves its historical failures. The self-scored harm endpoint remains non-authoritative. In a certificate-independent finite panel, authentication rejects four registered corruption types without unsafe reuse and at lower valid-panel cost than always raw. In the larger composed finite world, it rejects all 2,457 scheduled corruptions; removing authentication leaves 330 unsafe reuses despite similar aggregate correctness. Confidence and provenance remain useful signals, but neither specifies what distinctions state supports.

A responsibility/recovery contract can state the exact conditions under which compact-state reuse is authorized; independently graded external outcomes are still required to establish real-system safety.

9 References

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